Formoterol Neutec 12mcg Capsair Hard Capsule Inhalation Powder

Each inhalation powder capsules contains 12 mcg Formoterol Fumarate Dihydrate

Product Description

Cover; transparent
Body; in natural transparent capsule, white powder

Pharmacodynamics

Pharmacotherapeutic group: Selective beta2-adrenergic agonist, ATC code: R03AC13.
Formoterol is a potent selective β2-adrenergic stimulant. It exerts a bronchodilator effect in patients with reversible airways obstruction. The effect sets in rapidly (within 1-3 minutes) and is still significant 12 hours after inhalation.
Formoterol inhalation powder has been shown to be effective in preventing bronchospasm induced by exercise and methacholine. Formoterol has been shown to improve symptoms and pulmonary function and quality of life in the treatment of conditions associated with COPD. Formoterol acts on the reversible component of the disease.

Pharmacokinetics

Formoterol has a therapeutic dose range of 12 to 24 micrograms b.i.d. Data on the plasma pharmacokinetics of formoterol were collected after inhalation of doses higher than the recommended range and in COPD patients after inhalation of therapeutic doses. Urinary excretion of unchanged formoterol, used as an indirect measure of systemic exposure, correlates with plasma drug disposition data. The elimination half-lives calculated for urine and plasma are similar.

Absorption

Following inhalation of a single 120 microgram dose of formoterol fumarate, formoterol was rapidly absorbed into plasma, reaching a maximum concentration of 266 pmol/l within 5 minutes of inhalation. In COPD patients treated for 12 weeks with formoterol fumarate 12 or 24 micrograms b.i.d. the plasma concentrations of formoterol ranged between 11.5 and 25.7 pmol/L and 23.3 and 50.3 pmol/L respectively at 10 minutes, 2 hours and 6 hours post inhalation.

The cumulative urinary excretion of formoterol and/or its (R,R) and (S,S)-enantiomers, after inhalation of dry powder (12-96 micrograms) or aerosol formulations (12-96 micrograms), showed that absorption increased linearly with the dose.

After 12 weeks administration of 12 micrograms or 24 micrograms formoterol powder b.i.d., the urinary excretion of unchanged formoterol increased by 63-73% in adult patients with asthma, by 19-38% in adult patients with COPD and by 18-84% in children, suggesting a modest and self-limiting accumulation of formoterol in plasma after repeated dosing.

As reported for other inhaled drugs, it is likely that about 90% of formoterol administered from an inhaler will be swallowed and then absorbed from the gastrointestinal tract. This means that the pharmacokinetic characteristics of the oral formulation largely apply also to the inhalation powder. When 80 micrograms of ^{3}H-labelled formoterol fumarate was orally administered, at least 65% of the drug was absorbed.

<u>Distribution</u>

The plasma protein binding of formoterol is 61-64% (34% primarily to albumin).

There is no saturation of binding sites in the concentration range reached with therapeutic doses.

Biotransformation

Formoterol is eliminated primarily by metabolism, direct glucuronidation being the major pathway of biotransformation, with O-demethylation followed by further glucuronidation being another pathway. Minor pathways involve sulphate conjugation of formoterol and deformylation followed by sulphate conjugation. Multiple isozymes catalyze the glucuronidation (UGT1A1, 1A3, 1A6, 1A7, 1A8, 1A9, 1A10, 2B7 and 2B15) and O-demethylation (CYP2D6, 2C19, 2C9, and 2A6) of formoterol, and so consequently the potential for metabolic drug-drug interaction is low. Formoterol did not inhibit cytochrome P450 isozymes at therapeutically relevant concentrations. The kinetics of formoterol are similar after single and repeated administration, indicating no auto-induction or inhibition of metabolism.

Elimination

In asthmatic and COPD patients treated for 12 weeks with 12 or 24 micrograms formoterol fumarate b.i.d., approximately 10% and 7% of the dose, respectively, were recovered in the urine as unchanged formoterol. In asthmatic children, approximately 6% of the dose was recovered in the urine as unchanged formoterol after multiple dosing of 12 and 24 micrograms. The (R,R) and (S,S)-enantiomers accounted, respectively, for 40% and 60% of urinary recovery of unchanged formoterol, after single doses (12 to 120 micrograms) in healthy individuals and after single and repeated doses in asthma patients.

After a single oral dose of ^{3}H-formoterol, 59-62% of the dose was recovered in the urine and 32-34% in the faeces. Renal clearance of formoterol is 150 mL/min.

After inhalation, plasma formoterol kinetics and urinary excretion rate data in healthy individuals indicate a biphasic elimination, with the terminal elimination half-lives of the (R,R)- and (S,S)-enantiomers being 13.9 and 12.3 hours, respectively.

Approximately 6.4-8% of the dose was recovered in the urine as unchanged formoterol, with the (R,R) and (S,S)-enantiomers contributing 40% and 60% respectively.

Indications

- Prophylaxis and treatment of bronchoconstriction in patients with asthma.
- Prophylaxis of bronchospasm induced by inhaled allergens, cold air, or exercise.
- Prophylaxis and treatment of bronchoconstriction in patients with reversible or irreversible chronic obstructive pulmonary disease (COPD) including chronic bronchitis and emphysema. Formoterol Neutec 12 mcg Capsair Inhalation Powder has been shown to improve quality of life in COPD patients.

Recommended Dose

Method of administration

For inhalation use in adults and in children 5 years of age and older.

Formoterol Neutec 12 mcg Capsair inhalation powder capsules should be used only with the inhaler provided in the pack.

To ensure proper administration of the drug, a physician or other health professional should:

- Show the patient how to use the inhaler.
- Dispense the capsule only together with the inhaler.
- Instruct the patient that the capsules are only for inhalation use and not to be swallowed (see section WARNINGS AND PRECAUTIONS).

Detailed handling instructions are included in the package leaflet.

It is important for the patient to understand that the gelatin capsule might fragment and small pieces of gelatin might reach the mouth or throat after inhalation. The tendency for this to happen is minimised by not piercing the capsule more than once. The capsule made of edible gelatin is not

harmful if ingested.
The capsules should be removed from the blister pack only immediately before use.

Dosage

Adults

Asthma

For regular maintenance therapy, 1 to 2 inhalation capsules (equivalent to 12 to 24 micrograms formoterol) twice daily.The maximum recommended maintenance dose is 48 micrograms per day. If required, an additional 1 to 2 capsules per day may be used for the relief of ordinary symptoms provided the recommended daily maximum dose of 48 micrograms per day is not exceeded. However, if the need for additional doses is more than occasional (e.g. more frequent than 2 days per week) medical advice should be sought and therapy reassessed, as this may indicate a worsening of the underlying condition. Formoterol Neutec 12 mcg Capsair should not be used to relieve the acute symptoms of an asthma attack. In the event of an acute attack, a short-acting beta2-agonist should be used (see section WARNINGS AND PRECAUTIONS).

Prophylaxis against exercise induced bronchospasm or before exposure to a known unavoidable allergen

The content of 1 inhalation capsule (12 micrograms) should be inhaled at least 15 minutes prior to exercise or exposure. In patients with a history of severe bronchospasm , 2 inhalation capsules (24 micrograms) may be necessary as prophylaxis.

Chronic obstructive pulmonary disease

For regular maintenance therapy, 1 to 2 inhalation capsules (12 to 24 micrograms) twice daily.

Pediatrics (Children aged 5 years or older)

Asthma

For regular maintenance therapy, 1 inhalation capsule (12 micrograms) twice daily.
For children 5-12 years of age, treatment with a combination product containing an inhaled corticosteroid and long-acting beta2-agonist (LABA) is recommended, except in cases where a separate inhaled corticosteroid and long-acting beta2-agonist are required (see Section WARNINGS AND PRECAUTIONS).
The maximum recommended dose is 24 micrograms per day.
Formoterol Neutec 12 mcg Capsair should not be used to relieve the acute symptoms of an asthma attack. In the event of an acute attack, a short-acting beta2-agonist should be used (see section WARNINGS AND PRECAUTIONS).

Prophylaxis against exercise-induced bronchospasm or before exposure to a known unavoidable allergen

The content of 1 inhalation capsule (12 micrograms) should be inhaled at least 15 minutes prior to exercise or exposure.

Formoterol Neutec 12 mcg Capsair is not recommended in children under 5 years of age.

Adults and children aged 5 years or older:

The bronchodilator effect of Formoterol Neutec 12 mcg Capsair is still significant 12 hours after inhalation. Therefore, in most cases, twice-daily maintenance therapy will control the bronchoconstriction associated with chronic conditions, both during the day and at night.

Special populations

Renal impairment

The pharmacokinetics of formoterol has not been studied in patients with renal impairment.

Hepatic impairment
The pharmacokinetics of formoterol has not been studied in patients with hepatic impairment.
Geriatrics (older than 65 years)
The pharmacokinetics of formoterol has not been studies in the elderly population. The available data from clinical trials performed in elderly patients do not suggest that the dosage should be different than in other adults

Contraindication
Known hypersensitivity to formoterol, to lactose (which contains small amount of milk proteins) or to any of the other excipients of Formoterol Neutec 12 mcg Capsair Inhalation.

Warning and Precautions
Asthma-related death
Formoterol fumarate dihydrate, the active ingredient of Formoterol Neutec 12 mcg Capsair, belongs to the class of long-acting beta2-adrenergic agonists (LABAs). With salmeterol, a different long-acting beta2-agonist, a higher rate of death due to asthma was observed in the patients treated with salmeterol (13/13,176). No study adequate to determine whether the rate of asthma-related death is increased with Formoterol Neutec 12 mcg Capsair has been conducted.

In the treatment of asthma
Formoterol Neutec 12 mcg Capsair should not be used (and is not sufficient) as the first treatment for asthma.

When treating patients with asthma, use Formoterol Neutec 12 mcg Capsair only as an add-on to an inhaled corticosteroid (ICS) for patients who are not adequately controlled on an ICS alone or whose disease severity clearly warrants initiation of treatment with both an ICS and a LABA.

Children up to the age of 6 years should not be treated with Formoterol Neutec 12 mcg Capsair as sufficient experience is not available for this group. For children 6-12 years of age, when treatment with an ICS and LABA is required, it is recommended to use a combination product, except in cases where a separate ICS and LABA are more appropriate.

Formoterol Neutec 12 mcg Capsair should not be used in conjunction with another LABA

Whenever Formoterol Neutec 12 mcg Capsair is prescribed, patients should be evaluated for the adequacy of the anti-inflammatory therapy they receive. Patients must be advised to continue taking anti-inflammatory therapy unchanged after the introduction of Formoterol Neutec 12 mcg Capsair, even when their symptoms improve.

The daily dose of Formoterol Neutec 12 mcg Capsair should not be increased beyond the maximum recommended dose (see Section 4.2).

Once asthma symptoms are controlled, consideration may be given to gradually reducing the dose of Formoterol Neutec 12 mcg Capsair. Regular review of patients as treatment is stepped down is important. The lowest effective dose of Foradil should be used.

Serious asthma-related adverse events and exacerbations may occur during treatment with Formoterol Neutec 12 mcg Capsair. Higher incidence of serious asthma exacerbations in patients who received Formoterol Neutec 12 mcg Capsair, particularly in patients 5-12 years of age.

Patients should be advised that if, after initiation of Formoterol Neutec 12 mcg Capsair, their symptoms persist, or if the number of doses of Formoterol Neutec 12 mcg Capsair required to control their symptoms increases, this usually indicates a worsening of the underlying condition. In these circumstances, they should be advised to continue treatment but to seek medical advice as soon as possible.

Patients should not be initiated on Formoterol Neutec 12 mcg Capsair or the dose increased during an acute severe asthma exacerbation, or if they have significantly worsening or acutely deteriorating asthma.

Formoterol Neutec 12 mcg Capsair must not be used to relieve acute asthma symptoms. In the event of an acute attack, a short-acting beta2-agonist should be used. Patients must be informed of the need to seek medical treatment immediately if their asthma deteriorates suddenly.

Concomitant conditions
Special care and supervision, with particular emphasis on dosage limits, is required in patients receiving Formoterol Neutec 12 mcg Capsair when the following conditions may exist:

Ischaemic heart disease, cardiac arrhythmias, especially third-degree atrioventricular block, severe cardiac decompensation, idiopathic subvalvular aortic stenosis, severe hypertension, aneurysm, phaeochromocytoma, hypertrophic obstructive cardiomyopathy, thyrotoxicosis, or other severe cardiovascular disorders, such as tachyarrhythmias or severe heart failure.

Formoterol may induce prolongation of the QTc-interval. Caution should be observed when treating patients with prolongation of the QTc-interval and in patients treated with drugs affecting the QTc-interval.

Caution should be used when co-administering theophylline and formoterol in patients with pre-existing cardiac conditions

Due to the hyperglycaemic effect of β2-stimulants, including formoterol, additional blood glucose controls are recommended in diabetic patients.

Hypokalaemia
Potentially serious hypokalaemia may result from β2-agonist therapy, including formoterol. Particular caution is advised in severe asthma as this effect may be potentiated by hypoxia and concomitant treatment. It is recommended that serum potassium levels be monitored in such situations.

Paradoxical bronchospasm
As with other inhalation therapy, the potential for paradoxical bronchospasm should be kept in mind. If it occurs, the preparation should be discontinued immediately and alternative therapy substituted.

Formoterol Neutec 12 mcg Capsair contains lactose monohydrate less than 500 micrograms per delivered dose. This amount does not normally cause problems in lactose intolerant patients. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Incorrect route of administration

There have been reports of patients who have mistakenly swallowed Formoterol Neutec 12 mcg Capsair capsules instead of placing the capsules in the inhalation device. The majority of these ingestions were not associated with side effects. Healthcare providers should discuss with the patient how to correctly use Formoterol Neutec 12 mcg Capsair. If a patient who is prescribed Formoterol Neutec 12 mcg Capsair does not experience breathing improvement, the healthcare provider should ask how the patient is using Formoterol Neutec 12 mcg Capsair.

Interactions with Other Medicaments No specific interaction studies have been carried out with Formoterol Neutec 12 mcg Capsair.

There is a theoretical risk that concomitant treatment with other drugs known to prolong the QTc-interval may give rise to a pharmacodynamic interaction with formoterol and increase the possible risk of ventricular arrhythmias. Examples of such drugs include certain antihistamines (e.g. terfenadine, astemizole, mizolastine), certain antiarrhythmics (e.g. quinidine, disopyramide, procainamide), erythromycin and tricyclic antidepressants. In addition, levodopa, levothyroxine, oxytocin and alcohol can impair cardiac tolerance towards beta2-agonists.

Concomitant administration of other sympathomimetic agents such as other beta2 –agonists or ephedrine has the potential to produce additive effects both in respect of the desirable effects and the undesirable effects of Formoterol Neutec 12 mcg Capsair. Therefore, titration of the dose may be required.

Concomitant treatment with xanthine derivatives, steroids, or diuretics such as thiazides and loop diuretics may potentiate a possible hypokalaemic adverse effect of beta2-agonists. Hypokalaemia may increase susceptibility to cardiac arrhythmias in patients treated with digitalis glycosides.

There is an elevated risk of arrhythmias in patients receiving concomitant anaesthesia with halogenated hydrocarbons.

Formoterol may interact with monoamine oxidase inhibitors and should not be given to patients receiving treatment with monoamine oxidase inhibitors or for up to 14 days after their discontinuation.

Concomitant administration of formoterol and corticosteroids may increase the hyperglycaemic effect seen with these drugs.

The bronchodilating effects of formoterol can be enhanced by anticholinergic drugs.

Beta-adrenergic blockers may weaken or antagonise the effect of Formoterol Neutec 12 mcg Capsair. Therefore, Formoterol Neutec 12 mcg Capsair should not be given together with beta-adrenergic blockers (including eye drops) unless there are compelling reasons for their use.

Pregnancy and Lactation

Pregnancy

There were no teratogenic effects revealed in animal tests. In animal studies formoterol has caused implantation losses as well as decreased early postnatal survival and birth weight. The effects appeared at considerably higher systemic exposures than those reached during clinical use of formoterol. However, until further experience is gained, Formoterol Neutec 12 mcg Capsair is not recommended for use during pregnancy (particularly at the end of pregnancy or during labour) unless there is no more established alternative. As with any medicine, use during pregnancy should only be considered if the expected benefit to the mother is greater than any risk to the foetus.

Breastfeeding
The substance has been detected in the milk of lactating rats, but it is not known whether formoterol passes into human breast milk, therefore mothers using Formoterol Neutec 12 mcg Capsair should refrain from breast feeding their infants.

Fertility
There is no available data on the effect of formoterol on human fertility. No impairment of fertility was observed in studies performed in male and female rats

Side Effects

The most commonly reported adverse events of beta$_2$-agonist therapy, such as tremor and palpitations, tend to be mild and disappear within a few days of treatment**.**

Immune system disorders	Rare	Hypersensitivity reactions such as bronchospasm, severe hypotension, urticaria, angioedema, pruritus, exanthema, peripheral oedema
Metabolism and nutrition disorders	Rare	Hypokalaemia
	Very rare	Hyperglycaemia
Psychiatric disorders	Uncommon	Agitation, restlessness, sleep disturbances, anxiety
Nervous system disorders	Common	Headache, tremor
	Very rare	Dizziness, taste disturbance
Cardiac disorders	Common	Palpitations
	Uncommon	Tachycardia
	Rare	Cardiac arrhythmias, e.g atrial fibrillation, supraventricular tachycardia, extrasystoles
	Very rare	Angina pectoris, prolongation of QTc interval
Vascular disorders	Very rare	Variation in blood pressure
Respiratory, thoracic and mediastinal disorders	Rare	Aggravated bronchospasm, paradoxical bronchospasm (see Section 4.4), oropharyngeal irritation
Gastrointestinal disorders	Rare	Nausea
Musculoskeletal, connective tissue and bone disorders	Uncommon	Muscle cramps, myalgia

As with all inhalation therapy, paradoxical bronchospasm may occur in very rare cases.

Treatment with beta2-agonists may result in an increase in blood levels of insulin, free fatty acids, glycerol and ketone bodies.

Lactose monohydrate contains small amounts of milk proteins and can therefore cause allergic reactions.

Symptoms and Treatment of Overdose

<u>Symptoms</u>

There is limited clinical experience to date on the management of overdose, however, an overdosage of , Formoterol Neutec 12 mcg Capsair Inhalation Powder would be likely to lead to effects that are typical of β2-adrenergic agonists: nausea, vomiting, headache, tremor, somnolence, palpitations, tachycardia, ventricular arrhythmias, metabolic acidosis, hypokalaemia, hyperglycaemia, prolonged QTcinterval, hypertension.

<u>Treatment</u>

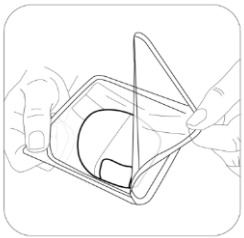
Supportive and symptomatic treatment is indicated. Serious cases should be hospitalised.

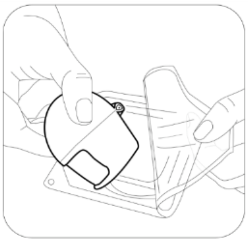
Use of cardioselective beta-blockers may be considered, but only subject to extreme caution since the use of β-adrenergic blocker medication may provoke bronchospasm. Serum potassium should be monitored.

Effects on Ability to Drive and Use Machine
Patients experiencing dizziness or other similar side effects should be advised to refrain from driving or using machines

Instruction for Use
The inhalation device is for breathing (inhaling) the medicine inside the Formoterol Neutec inhaler capsule, prescribed by your doctor for breathing problems.
Inhaler device which is launched in the cardboard within the protective packaging for safety purposes.

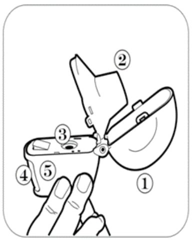
Please take out your device from the package before using as shown at figure.



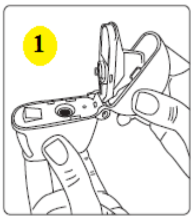


While using Formoterol Neutec, please do not forget to follow the instructions of your doctor carefully. The inhaler device is specifically designed for inhalation of Formoterol Neutec, inhalation capsule. It should not be used to take another drug.

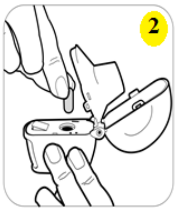
1. Powder cap
2. Mouthpiece
3. Central reservoir
4. Drill button
5. Bottom reservoir



1. Pulling the dust head up ensure that the mouthpiece is fully opened.



2. Take a Formoterol Neutec inhalation capsule from blister package (It is important that you take the capsule from the blister pack only immediately before you use it.) and as shown at figure 2, place it in the central reservoir. It does not matter direction of the capsule in the reservoir.



3. Close the mouthpiece tightly until it clicks. Leave the powder cap open.

4. Keep the inhaler device upright and push the drilling button only a single motion and leave. Thus holes will be opened in the capsule and will allow the drug to be released when you breathe.

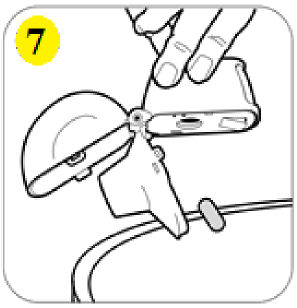
5. Breathe out fully.

Important: please never breathe into the mouthpiece.

6. Lift the inhalation device up to your mouth and close your lips tightly around the mouthpiece. Take a slow and deep breath by holding your head upright; adjust your breathing rate to hear or feel the capsule vibrating.

Inhale until your lungs are full; hold your breath for a period of time and remove the inhalation device from your mouth**.** Then breathe normally. Repeat steps 5 and 6 to empty capsule.

7. Open the mouthpiece again. Turn the device and throw the used capsule. To clean powders in the inside, use a dry soft tissue or a clean soft brush. Close the mouthpiece and powder cap and leave your device.



NOTE: **DO NOT USE WATER** to clean the inhaler device. The capsules should not be exposed to extreme temperatures. Formoterol Neutec inhaler capsules contain a small amount of powder, so capsules are only partially filled.

Storage Condition

Store at room temperature below 30°C.

Shelf life
15 months

Dosage form and packaging available
The Formoterol Neutec 12 mcg Capsair Inhalation Powder pack includes a blister pack of 60 capsules containing the medicine and an inhaler device for taking the medicine.

Manufacturer
Neutec İnhaler İlaç Sanayi ve Ticaret. A.Ş
Turkey, Sakarya, Arifiye, 1. Organize Sanayi Bölgesi,
2 Nolu Yol, No:3

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